BRAC University

Department of Mathematics and Natural Sciences

Lecture on

Real Analysis (MAT221)

Fundamentals of Calculus

History of Calculus, Fundamental Theorem of Calculus

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Conducted by

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History of Calculus

Fundamental Theorem of Calculus

Fundamental Theorem of Calculus Part 1

Let f be a continuous real-valued function defined on a closed interval $[a, b]$. Define the function F on $[a, b]$ by

$$F(x) = \int_a^x f(t)\, dt.$$

Then, F is continuous on $[a, b]$, differentiable on (a, b), and

$$F'(x) = f(x) \quad \text{for all } x \in (a, b).$$

Fundamental Theorem of Calculus Part 2

Let f be a real-valued function defined on a closed interval $[a, b]$ that is Riemann integrable. If F is any antiderivative of f on $[a, b]$, then

$$\int_a^b f(x)\,dx = F(b) - F(a).$$

Some Consequences of the Fundamental Theorem of Calculus

Theorem

We have seen that not every derivative is continuous, but explain how we at least know that every continuous function is a derivative.

Theorem

Let

$$f(x) = |x|$$

and define

$$F(x) = \int_{-1}^{x} f(t)\,dt.$$

1. Find a formula for $F(x)$ for all x. Where is F continuous? Where is F differentiable? Where does

$$F'(x) = f(x)?$$

2. Repeat part (1) for the function

$$f(x) = \begin{cases} 1, & \text{if } x < 0, \\ 2, & \text{if } x \geq 0. \end{cases}$$

Theorem

The hypothesis in FTC-1 that

$$F'(x) = f(x)$$

for all $x \in [a, b]$ is slightly stronger than it needs to be. Carefully read the proof and state exactly what needs to be assumed with regard to the relationship between f and F for the proof to be valid.

Average Value Theorem

If g is continuous on $[a, b]$, show that there exists a point

$$c \in (a, b)$$

such that

$$g(c) = \frac{1}{b-a} \int_a^b g(x)\, dx.$$

Theorem

Let

$$h(x) = \begin{cases} 1, & \text{if } x < 1 \text{ or } x > 1, \\ 0, & \text{if } x = 1, \end{cases}$$

and define

$$H(x) = \int_0^x h(t)\,dt.$$

Show that even though h is not continuous at $x = 1$, $H(x)$ is still differentiable at $x = 1$.

Thank You!

We'd love your questions and feedback.

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References

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[2] Terence Tao, *Analysis I*, 3rd Edition, Texts and Readings in Mathematics, Hindustan Book Agency, 2016.